EMSCNet: Efficient Multisample Contrastive Network for Remote Sensing Image Scene Classification

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***Abstract*— Significant progress has been achieved in remote sensing image scene classification (RSISC) with the development of convolutional neural networks (CNNs) and vision transformers (ViTs). However, high intraclass diversity and interclass similarity are still enormous challenges for RSISC. Metric learning can effectively improve the discriminative ability of deep representations by constraining the distance between features. Previous metric learning methods only optimize the feature space representation through metric function, ignoring the information interaction between samples. For complex scene images, similarity and discriminative knowledge need to be summarized from the multiple positive and negative pairs. We propose a novel efficient multisample contrastive network (EMSCNet) to integrate knowledge from multiple samples. Specifically, we construct a dynamic dictionary with momentum updates to mine positive and negative pairs from the entire dataset. Then, the similarity and discriminative knowledge between samples are summarized by introducing a contrastive module. Finally, the knowledge of the contrastive module is transferred to the backbone classifier through knowledge distillation. The proposed contrastive module can be easily embedded into the training process of CNNs or ViT and removed during inference. Experimental results conducted on three datasets demonstrate the effectiveness of the proposed approach.**

***Index Terms*— Convolutional neural networks (CNNs), knowledge distillation, metric learning, remote sensing image scene classification (RSISC), vision transformer (ViT).**

I. Introduction

IN RECENT years, with the development of high-resolution satellite imaging instruments, remote sensing technology has become a meaningful way to explore the state of the earth's surface. Remote sensing images are widely used in remote sensing image scene classification (RSISC) [1], [2], [3], [4], image fusion [5], [6], and change detection [7]. RSISC is a fundamental task in remote sensing image analysis, aiming to assign a unique semantic label for scene images based on their visual information. In general, these scene image has the characteristics of broad independent coverage, high resolution, and coexistence of multiple ground objects, which is quite different from the natural images. The problems of high intraclass diversity and high interclass similarity are two major challenges for scene classification [8]. Specifically, there are two main reasons for intraclass diversity. On the one hand, the same ground object has multiple forms due to different colors, shapes, and scales. As shown in Fig. 1(a), the color and appearance of buildings in the industrial category show great differences, and the building density and vegetation coverage in Fig. 1(b) lead to different architectural styles in the school category.On the other hand, the scene image may contain some noise due to the weather, clouds, and other natural factors. As shown in Fig. 1(c), the scene images taken in cloudy and foggy weather are noticeably more blurry. The main reason for the high interclass similarity is that there may be high semantic overlap between different categories due to the wide coverage



Fig. 1. Main challenges for RSISC include (a)–(c) high intraclass diversity and (d)–(f) high interclass similarity.

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of scene images. As shown in Fig. 1(d) and (e), the *bridge* and *overpass* categories both contain the bridge building, and the *playground* and *stadium* both contain the race track. In addition, the distribution density of ground objects also leads to images with the same components but different semantic labels. As shown in Fig. 1(f), both *dense residential* and *medium residential* contain houses, trees, grass, etc. The distribution density of houses leads to inconsistent semantic labels.

Benefiting from the powerful feature extraction capability of convolutional neural networks (CNNs), remarkable progress has been made in computer vision in the past decade [9]. However, due to the complex composition of scene images, even CNNs cannot accurately distinguish the high intraclass diversity and high interclass similarity images. To address the above challenges, some approaches [10], [11], [12], [13] optimize the interclass and the intraclass distances through metric learning. Although these methods improve the problem of category confusion between scene images, directly applying the distance metric loss may lead to the overcorrection of features [14]. Humans typically observe and contrast images with high intraclass diversity to summarize their similarities and try to find differences in images with high interclass similarity to distinguish them. If simply restricting the sample similarity by a distance function, the network may have difficulty in summarizing effective similarity and discriminative knowledge since metric learning neglects the information interaction between samples. In addition, metric learning methods create positive and negative pairs in a mini-batch, resulting in the number of contrastive samples being limited by the mini-batch size. Meanwhile, generating positive pairs and mining hard pairs is difficult when the number of categories is larger than the mini-batch size. It is impractical to use large batch size due to equipment limitations.

To address the above challenges, this article proposes an efficient multisample contrastive network (EMSCNet), which provides a new perspective on the traditional metric learning typically used in RSISC tasks [10], [11]. Specifically, we propose a contrastive module to integrate knowledge between samples, which consists of a sample dictionary and a multihead contrastive block. Samples taken from the same category are considered as positive pairs, and those taken from different categories are considered as negative pairs. Instead of correcting sample space distance only by a metric function, the multihead contrastive block can effectively extract discriminative and similarity knowledge from positive and negative pairs by the information interaction method. Further, we address the mini-batch size limitation by a dictionary construction method [15]. The advantage of this approach is that the dictionary size and mini-batch size are decoupled, allowing it to be large. To preserve the consistency of the features in the dictionary, i.e., the features stored in the dictionary should come from encoders with similar parameters, we design a momentum encoder that slowly updates parameters via the momentum method to assist dictionary updating. In addition, we introduce a multisample contrastive loss [16] to assist the learning of the contrastive module, which can optimize the spatial distance of multiple positive and negative pairs at the same time. Finally, the dark knowledge of the contrastive module is transferred to the backbone classifier via knowledge distillation. The main contributions of this article are summarized as follows.

1) To address the challenges of high intraclass diversity and high interclass similarity for RSISC, we propose an EMSCNet by extending the traditional metric learning and unsupervised contrastive learning.
2) We propose a contrastive module integrating a dynamic dictionary, which can effectively summarize the similarity and discriminative knowledge from positive and negative sample pairs through the information interaction method. Meanwhile, we introduce a multisample contrastive function to assist the learning of it.
3) We transfer the dark knowledge of the contrastive module to the backbone classifier through knowledge distillation. Thus, the contrastive module can be removed during inference, and it does not introduce additional burden.

The rest of this article is organized as follows. Section II introduces the progress of RSISC methods and the metric learning method closely related to the method proposed in this article. In Section III, the proposed EMSCNet model is described in detail. Section IV presents detailed experimental results on three datasets and analyzes the results. Finally, we conclude our work in Section V.

II. Related Work

A. Remote Sensing Image Scene Classification

1) Handcrafted Features Methods: The early handcrafted feature-based methods mainly extracted color, texture, and edges from images as features [17], [18], [19]. Typical handcrafted feature extraction methods are texture descriptors (TD) [20], scale-invariant feature transformation (SIFT) [21], color histogram (CH) [22], and histogram of oriented gradients (HOG) [23]. CH and TD can be directly used for scene classification by extracting global features of images. SIFT and HOG are usually combined with mid-level descriptors [1], [24] to extract global features. Zhu et al. [25] proposed a bag-of-visual-words model for scene classification, which can combine the local and global features by feature fusion. Handcrafted feature methods rely on manual design, and it is difficult to meet the requirements of complex scene image analysis.

2) CNNs- and ViT-Based Methods: With the rise of the deep learning wave, the great potential of CNNs methods to extract high-level features has attracted the attention of the remote sensing community [26], [27]. Li et al. [28] proposed a key filter bank that can extract local key information while preserving global information, which fully integrates local and global features. Chen et al. [29] proposed a multibranch local attention network to extract features by attention mechanism, and thus effectively enhancing the feature representation ability of the network. Bai et al. [30] presented to extract the multifrequency and multiscale features from scene images and then used global covariance pooling instead of global average pooling, which effectively captured the richer statistical information

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of deep features. Zhao et al. [31] proposed a feature fusion branch to extract multigranularity and multilevel features, and then fused multiple prediction results by the ensemble learning method.

Due to the complexity of the scene images, establishing long-range dependencies is beneficial for scene classification. CNNs obtains an enlarged receptive field by stacking convolutional layers, where local elements interact indirectly. Therefore, it is difficult for CNNs to capture long-range contextual relations. The transformer [32] can establish the long-range dependencies between local elements through a self-attention mechanism. ViT [33] introduced transformer into the visual field and achieved superior performance. Bazi et al. [34] demonstrated the ViT-based method has great potential in RSISC. Tang et al. [35] combined the advantages of convolution and transformer to propose a contextual information extraction module, which can capture rich contextual information from multilevel features. Lv et al. [36] proposed to preserve local structural features by aggregating the neighboring overlapping patches, and consider the importance of different channels through a lightweight channel attention module. Xu et al. [37] transfer the dark knowledge from the teacher ViT network to the student ResNet network through knowledge distillation, so that the student can achieve similar accuracy as teacher. Although these methods achieve advanced results, they ignore the potential relationship between samples.

3) Deep Metric Learning Methods: Deep metric learning [38], [39], [40], [41], [42] aims to learn a embedding function that maps the input data space to a low-dimensional embedding space. In the new space, the positive sample pairs are kept closer, and the negative sample pairs are pushed farther apart. Recent work has pointed out that metric learning methods can effectively represent aerial scene images [26]. Cheng et al. [10] proposed to impose a metric learning regularization term to enforce the model more discriminative. Wang et al. [14] presented to capture salient features through attention mechanism, and add triple metric regularization to constrain the distance, which can avoid the excessive feature correction caused by only using metric learning. Cheng et al. [43] integrated metric learning into few-shot learning and proposed to utilize a certain distance metric between the prototypes and the support features to classify support samples. Zhang et al. [12] designed a branch for the triplet loss and used dropout technology to improve the generalization ability of the model. Due to the complex semantics of remote sensing images, it may not be optimal to apply metric learning to remote sensing images directly. Kang et al. [13] proposed to discover the neighborhood structure in the metric space through scalable neighborhood component analysis, while preserving the model's discriminative ability with cross-entropy loss. However, metric learning only optimizes the distance of representations through a metric function and ignores the information interaction between sample pairs.

4) Contrastive Learning Methods: Contrastive learning extends deep metric learning to the field of unsupervised representation learning and has achieved exciting results [44], [45], [46], [47]. The ability of the network to mine positive and negative pairs is limited by mini-batch. Wu et al. [48] maintain a memory bank to store feature vectors, enabling fast nearest neighbor retrieval at the run time. MoCo [15] proposed to view contrastive learning as a dictionary look-up process and construct a dynamic dictionary to decouple the dictionary size from the mini-batch size. However, such storage methods are rarely used in supervised learning. We explore the effectiveness of dictionary storage methods for RSISC and obtain insightful results.

B. Deep Metric Learning Loss Function

The loss function of metric learning is mainly based on contrastive loss [49] and triplet loss [50]. We define the dataset as $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]^T$ with N images, and $\boldsymbol{y} = [y_1, y_2, \ldots, y_N]^T$ is the label vector. Definition $\boldsymbol{v}_i$ is the feature vector of sample $\boldsymbol{x}_i$ after network encoding. The Dis(., .) is the distance function, Max(., .) is the max function, and m is a margin for better similarity separation. The purpose of contrastive loss is to reduce the distance between samples of the same category and increase the distance between samples of different categories. It can be formulated as follows:

$$L_{\text{contrastive}} = \sum_{i,j}\Big[h_{i,j}\text{Dis}\big(\boldsymbol{v}_i, \boldsymbol{v}_j\big)^2 + \big(1 - h_{i,j}\big) \times \text{Max}\big(0, m - \text{Dis}\big(\boldsymbol{v}_i, \boldsymbol{v}_j\big)\big)^2\Big] \quad (1)$$

$$h_{i,j} = \begin{cases} 1, & \text{if } y_i = y_j \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The triple loss aims to optimize the distance from the positive samples to the anchor to be smaller than the distance from the negative samples to the anchor. It can be formulated as follows:

$$L_{\text{triplet}} = \sum_{i,j,k}\text{Max}\big(0, \text{Dis}\big(\boldsymbol{v}_i, \boldsymbol{v}_j\big) - \text{Dis}(\boldsymbol{v}_i, \boldsymbol{v}_k) + m\big) \quad (3)$$

where $\boldsymbol{v}_j$ and $\boldsymbol{v}_k$ are the positive and negative sample features of $\boldsymbol{v}_i$, respectively. However, these metric losses can only be optimized for one sample pair at a time. Sun et al. [16] proposed a circle loss to alleviate the problem of equal penalty intensity for every single similarity score, and it can optimize multiple positive and negative pairs at the same time. The circle loss can be formulated as follows:

$$L_{\text{circle}} = \sum_i\left[\log\left(1 + \sum_k\sum_j \exp\Big(\gamma\Big(\alpha_i^j\text{Dis}\big(\boldsymbol{v}_i, \boldsymbol{v}_j\big) - \alpha_i^k\text{Dis}(\boldsymbol{v}_i, \boldsymbol{v}_k)\Big)\Big)\right)\right] \quad (4)$$

where α_i^j and α_i^k are non-negative weighting factors, γ is a scale factor.

III. Method

The flowchart of EMSCNet is shown in Fig. 2. For the training set $\boldsymbol{X}_{\text{train}} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]^T$ with N images, the feature vector $\boldsymbol{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_N]^T$ is obtained through

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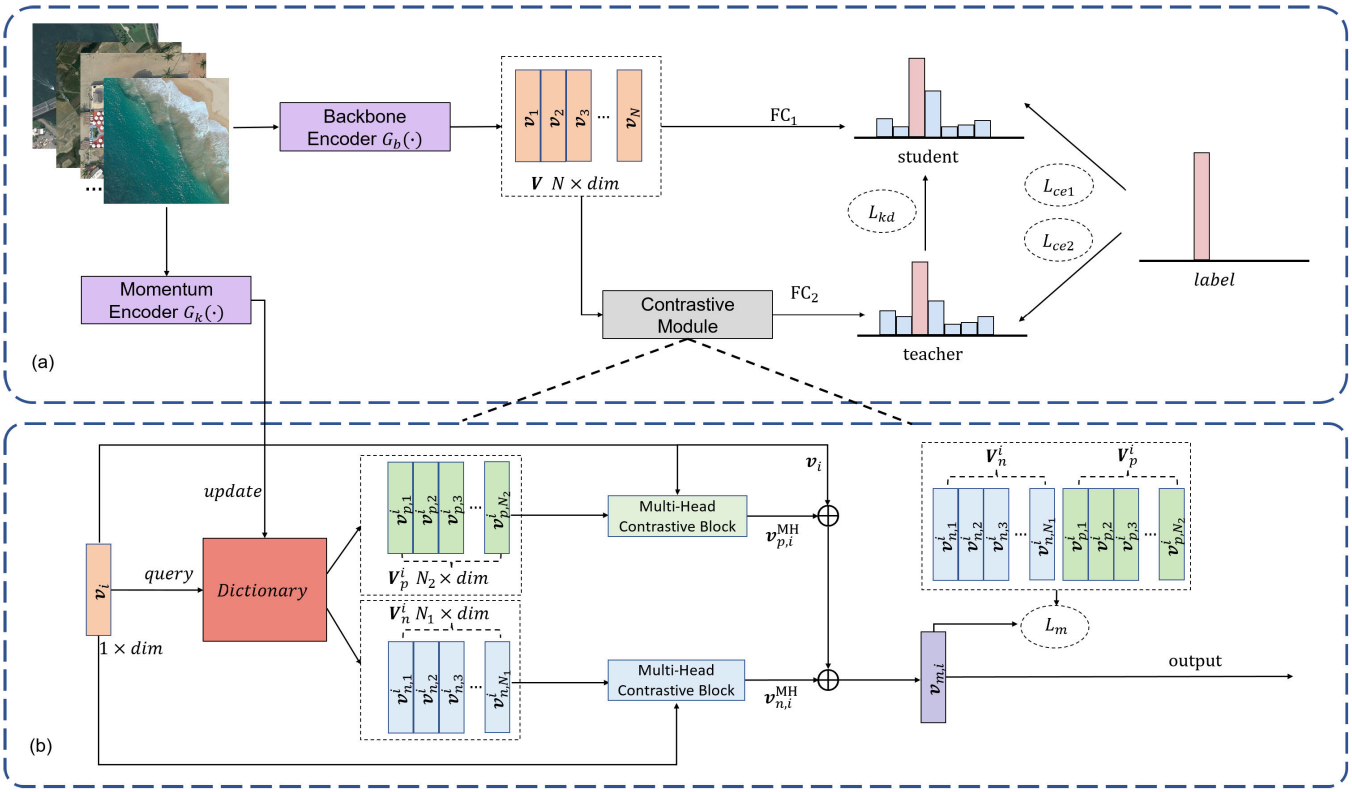


Fig. 2. (a) Framework of the proposed EMSCNet. (b) Contrastive module. For brevity, we assume that the number of feature vector is 1 in (b). (L_{ce1}: cross-entropy loss of backbone encoder, L_{ce2}: cross-entropy loss of contrastive module, L_m: multisample contrastive loss, L_{kd}: knowledge distillation loss, and $\oplus$: matrix addition.)

the backbone encoder $G_b(\cdot)$. The backbone network can use ViT [33] or ResNet [9], and ViT-B is used by default. The contrastive module takes each feature vector $\boldsymbol{v}_i$ as input, and outputs the feature vector $\boldsymbol{v}_{m,i}$. It can aggregate similarity and discriminative knowledge from positive and negative pairs. In the end of EMSCNet, the dark knowledge of the contrastive module is transferred to the backbone network by knowledge distillation. Therefore, in the inference stage, the contrastive module can be removed. In Section III-A, we introduce the dictionary construction method. Details of the multihead contrastive block are provided in Section III-B. In Section III-C, we illustrate the details of the loss function.

A. Dictionary Construction

Generally, the high-level semantic information of the image is summarized in the last layer of the network. We propose a contrastive module to integrate the information of positive and negative pairs. It can easily be embedded before the last fully connected (FC) layer to aggregate the high-level information. As shown in Fig. 2(b), for the feature vector $\boldsymbol{v}_i$ encoded by the backbone, we query from the dictionary to obtain a set of positive samples $\boldsymbol{V}_n^i = [\boldsymbol{v}_{n,1}^i, \boldsymbol{v}_{n,2}^i, \ldots, \boldsymbol{v}_{n,N_1}^i]^T$ and a set of negative samples $\boldsymbol{V}_p^i = [\boldsymbol{v}_{p,1}^i, \boldsymbol{v}_{p,2}^i, \ldots, \boldsymbol{v}_{p,N_2}^i]^T$, with N_1 and N_2 being the numbers of negative and positive samples, respectively. We introduce a dictionary construction method [15] to preserve feature consistency within each training epoch. Specifically, for a train set with M categories, we assign an array for each category to store the feature vectors and take the set of M arrays as a dictionary. The dictionary contains both query and update operations.

1) Dictionary Query: We query positive and negative samples from dictionary instead of mini-batch, which allows us to construct positive and negative pairs from the entire dataset. We take the same category samples as positive samples and different category samples as negative samples. For each query operation, the $k_1\%$ samples randomly selected from the same category array are taken as the positive samples, and the $k_2\%$ samples randomly selected from other category arrays are taken as the negative samples.

2) Dictionary Update: We expect the features stored in the dictionary to be obtained by the encoder $G_b(\cdot)$ with up-to-date parameters. However, this condition is difficult to satisfy because the network training mechanism is updated according to the mini-batch. This means we need to update the entire dictionary after each mini-batch training. We introduce a momentum update method [15] to improve the above problems, and initialize an encoder $G_k(\cdot)$ that is consistent with the structure of $G_b(\cdot)$. $G_k(\cdot)$ is updated with momentum m. The momentum update can be formulated as follows:

$$\theta_k^i = m\theta_k^{i-1} + (1-m)\theta_b^i \tag{5}$$

where θ_k^i and θ_b^i are the parameters of $G_k(\cdot)$ and $G_b(\cdot)$ in the ith epoch. $m \in [0, 1)$ is the momentum coefficient. As m increases, the encoder $G_k(\cdot)$ is updated more slowly. Therefore, the features in the array can be treated as taking from the encoders with similar parameters, which can preserve the consistency of features among all the mini-batch at each

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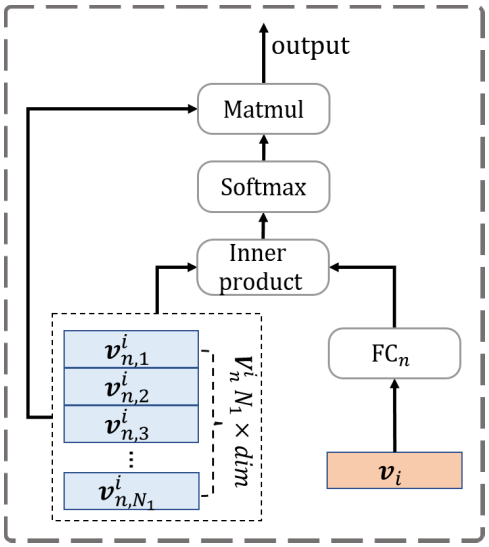


Fig. 3. Single-head contrastive block.

epoch. It is worth noting that only the encoder $G_b(\cdot)$ with θ^i_b is updated by means of backpropagation. For each mini-batch, we store the feature vectors encoded by the encoder $G_k(\cdot)$ into the array according to their labels, and then remove the oldest element in the array to keep the consistency with the current features.

B. Multihead Contrastive Block

The multihead contrastive block $\mathrm{MH}(\cdot,\cdot)$ is used to integrate similarity and discriminative knowledge from positive and negative sample pairs. It comprises several single-head contrastive block $\mathrm{SH}(\cdot,\cdot)$. Fig. 3 shows the flow of feature vector $\boldsymbol{v}_i$ integrating negative samples $\boldsymbol{V}^i_n$ knowledge in $\mathrm{SH}(\cdot,\cdot)$. The momentum update can stabilize the consistency between feature vectors in the dictionary and optimize $G_k(\cdot)$ toward $G_b(\cdot)$. However, there is still a semantic gap between $G_b(\cdot)$ and $G_k(\cdot)$, so we add an FC layer for semantic alignment. The aligned feature $\boldsymbol{v}'_{n,i}$ can be formulated as follows:

$$\boldsymbol{v}'_{n,i} = \mathrm{FC}_n(\boldsymbol{v}_i). \tag{6}$$

We calculate contrastive information via inner-product and it can be formulated as follows:

$$\begin{aligned}\boldsymbol{S}^i_n &= \sigma\left(\left\langle \boldsymbol{v}'_{n,i}, \boldsymbol{v}^i_{n,1}\right\rangle, \left\langle \boldsymbol{v}'_{n,i}, \boldsymbol{v}^i_{n,2}\right\rangle, \ldots, \left\langle \boldsymbol{v}'_{n,i}, \boldsymbol{v}^i_{n,N_1}\right\rangle\right) \\ &= \left[s^i_{n,1}, s^i_{n,2}, \ldots, s^i_{n,N_1}\right]\end{aligned} \tag{7}$$

where $\langle\cdot,\cdot\rangle$ is the inner-product, and $\sigma(\cdot)$ is the softmax function. The $s^i_{n,j}$ represents the response value of the jth negative sample to the ith sample, which controls the amount of contrastive information propagated from the jth negative sample to the ith sample. The $s^i_{n,j}$ is multiplied by the feature vector $\boldsymbol{v}^i_{n,j}$ to obtain the contrastive knowledge from the jth negative sample, and then the output of $\mathrm{SH}(\cdot,\cdot)$ is obtained by summing all negative samples' contrastive knowledge, i.e.,

$$\boldsymbol{v}^{\mathrm{SH}}_{n,i} = \mathrm{SH}\left(\boldsymbol{v}_i, \boldsymbol{V}^i_n\right) = \sum_{j=1}^{N_1} s^i_{n,j} \cdot \boldsymbol{v}^i_{n,j}. \tag{8}$$

Furthermore, we divide $\boldsymbol{v}_i$ and $\boldsymbol{V}^i_n$ into consecutive sequences along the channel, and feed them into the $\mathrm{SH}(\cdot,\cdot)$. Subsequently, the outputs of $\mathrm{SH}(\cdot,\cdot)$ are concatenated along the channel as the output of $\mathrm{MH}(\cdot,\cdot)$. The $\mathrm{MH}(\cdot,\cdot)$ allows for contrastive information between samples from different subspaces. It can be formulated as follows:

$$\begin{aligned}\boldsymbol{v}^{\mathrm{MH}}_{n,i} &= \mathrm{MH}\left(\boldsymbol{v}_i, \boldsymbol{V}^i_n\right) \\ &= \mathrm{Concat}\left(\mathrm{SH}_1\left(\boldsymbol{v}_{i,1}, \boldsymbol{V}^{i,1}_n\right), \ldots, \mathrm{SH}_T\left(\boldsymbol{v}_{i,T}, \boldsymbol{V}^{i,T}_n\right)\right)\end{aligned} \tag{9}$$

where T represents the number of heads, and T is set as 16 by default. $\mathrm{Concat}(\cdot)$ denotes concatenation operation, and $\mathrm{SH}_t(.,.)$ is the tth single-head contrastive block. Similarly, for the positive samples $\boldsymbol{V}^i_p$, we can get the output of $\mathrm{MH}(\cdot,\cdot)$ as follows:

$$\boldsymbol{v}'_{p,i} = \mathrm{FC}_p(\boldsymbol{v}_i) \tag{10}$$

$$\begin{aligned}\boldsymbol{S}^p_i &= \sigma\left(\left\langle \boldsymbol{v}'_{p,i}, \boldsymbol{v}^i_{p,1}\right\rangle, \left\langle \boldsymbol{v}'_{p,i}, \boldsymbol{v}^i_{p,2}\right\rangle, \ldots, \left\langle \boldsymbol{v}'_{p,i}, \boldsymbol{v}^i_{p,N_2}\right\rangle\right) \\ &= \left[s^i_{p,1}, s^i_{p,2}, \ldots, s^i_{p,N_2}\right]\end{aligned} \tag{11}$$

$$\boldsymbol{v}^{\mathrm{SH}}_{p,i} = \mathrm{SH}\left(\boldsymbol{v}_i, \boldsymbol{V}^i_p\right) = \sum_{j=1}^{N_2} s^i_{p,j} \cdot \boldsymbol{v}^i_{p,j} \tag{12}$$

$$\begin{aligned}\boldsymbol{v}^{\mathrm{MH}}_{p,i} &= \mathrm{MH}\left(\boldsymbol{v}_i, \boldsymbol{V}^i_p\right) \\ &= \mathrm{Concat}\left(\mathrm{SH}_1\left(\boldsymbol{v}_{i,1}, \boldsymbol{V}^{i,1}_p\right), \ldots, \mathrm{SH}_T\left(\boldsymbol{v}_{i,T}, \boldsymbol{V}^{i,T}_p\right)\right).\end{aligned} \tag{13}$$

We add $\boldsymbol{v}^{\mathrm{MH}}_{n,i}$, $\boldsymbol{v}^{\mathrm{MH}}_{p,i}$, and $\boldsymbol{v}_i$ to get the anchor feature vector $\boldsymbol{v}_{m,i}$ that integrates the contrastive information of positive and negative samples

$$\boldsymbol{v}_{m,i} = \boldsymbol{v}^{\mathrm{MH}}_{n,i} + \boldsymbol{v}^{\mathrm{MH}}_{p,i} + \boldsymbol{v}_i. \tag{14}$$

C. Loss Function

The loss function consists of three parts: cross-entropy loss, multisample contrastive loss, and knowledge distillation loss. The cross-entropy loss is to guide the network to learn the true labels. The multisample contrastive loss is to increase the interclass distance and reduce the intraclass distance. The knowledge distillation loss transfers the dark knowledge of the contrastive module to the backbone during the training stage.

1) Cross-Entropy Loss: The backbone network predictive vector $\mathrm{predict}_1$ can be obtained by performing an FC layer and a softmax function on $\boldsymbol{v}_i$. For the ith sample, the probability of class k can be formulated as follows:

$$\mathrm{predict}_1(k) = \frac{\exp\left(\mathrm{FC}_1(\boldsymbol{v}_i)^k\right)}{\sum_{j=1}^{M}\exp\left(\mathrm{FC}_1(\boldsymbol{v}_i)^j\right)}. \tag{15}$$

Similarly, the result of the contrastive module can be formulated as follows:

$$\mathrm{predict}_2(k) = \frac{\exp\left(\mathrm{FC}_2\left(\boldsymbol{v}_{m,i}\right)^k\right)}{\sum_{j=1}^{M}\exp\left(\mathrm{FC}_2\left(\boldsymbol{v}_{m,i}\right)^j\right)}. \tag{16}$$

We optimize these two classification results with cross-entropy loss. It can be formulated as follows:

$$L_{\mathrm{ce}_1} = -\sum_{k=1}^{M}\left(y_k \cdot \log\left(\mathrm{predict}_1(k)\right)\right) \tag{17}$$

$$L_{\mathrm{ce}_2} = -\sum_{k=1}^{M}\left(y_k \cdot \log\left(\mathrm{predict}_2(k)\right)\right). \tag{18}$$

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2) Multisample Contrastive Loss: Generally, metric learning loss is mainly based on contrastive loss and triplet loss. However, they cannot optimize multiple positive and negative pairs at the same time. We introduce a multisample contrastive loss [16], which can simultaneously optimize multiple positive and negative pairs. Guided by this loss, the contrastive module can further reduce the intraclass distance and increase the interclass distance. Denote d_n^i and d_p^j represent the cosine similarity distance from the ith negative and jth positive samples to the anchor, respectively. We expect the minimum similarity distance of positive pairs to be greater than the maximum similarity distance of negative pairs. The objective function can be formulated as follows:

$$L_m = \text{Max}\left(0, \gamma\left(\text{Max}\left(d_n^1, \ldots, d_n^{N_1}\right) - \text{Min}\left(d_p^1, \ldots, d_p^{N_2}\right) + m\right)\right) \tag{19}$$

where m is a margin for better similarity separation. For convenience, we add a coefficient γ. The relu function $\text{Max}(0, x)$ can be approximated by the soft-plus function. We approximated the max function $\text{Max}(.,.)$ and the min function $\text{Min}(.,.)$ with log-sum-exp function. It can be formulated as follows:

$$\text{Max}(0, x) \approx \log(1 + \exp(x)) \tag{20}$$

$$\text{Max}\left(d_n^1, \ldots, d_n^{N_1}\right) \approx \lim_{\gamma \to +\infty} \frac{1}{\gamma} \log\left[\sum_{i=1}^{N1} \exp\left(\gamma \cdot d_n^i\right)\right] \tag{21}$$

$$\begin{aligned}\text{Min}\left(d_p^1, \ldots, d_p^{N_2}\right) &= -\text{Max}\left(-d_p^1, \ldots, -d_p^{N_2}\right) \\ &\approx -\lim_{\gamma \to +\infty} \frac{1}{\gamma} \log\left[\sum_{i=1}^{N2} \exp\left(-\gamma \cdot d_p^i\right)\right].\end{aligned} \tag{22}$$

We can get the multisample contrastive loss L_m as follows:

$$\begin{aligned}L_m &\approx \lim_{\gamma \to +\infty} \log\left[1 + \exp\left(\log\left(\sum_{i=1}^{N1} \exp\left(\gamma\left(d_n^i + m\right)\right)\right)\right.\right. \\ &\qquad \left.\left. + \log\left(\sum_{j=1}^{N2} \exp\left(-\gamma \cdot d_p^j\right)\right)\right)\right] \\ &= \lim_{\gamma \to +\infty} \log\left[1 + \sum_{i=1}^{N_1} \exp\left(\gamma\left(d_n^i + m\right)\right) \sum_{j=1}^{N_2} \exp\left(-\gamma \cdot d_p^j\right)\right].\end{aligned} \tag{23}$$

We take derivatives of d_n^k and d_p^k as follows:

$$\frac{\partial L_m}{\partial d_n^k} = \frac{\gamma \cdot \exp\left(\gamma\left(d_n^k + m\right)\right) \sum_{j=1}^{N_2} \exp\left(-\gamma \cdot d_p^j\right)}{1 + \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \exp\left(\gamma\left(d_n^i + m\right) - \gamma \cdot d_p^j\right)} \tag{24}$$

$$\frac{\partial L_m}{\partial d_p^k} = \frac{-\gamma \cdot \exp\left(-\gamma \cdot d_p^k\right) \sum_{i=1}^{N_1} \exp\left(\gamma\left(d_n^i + m\right)\right)}{1 + \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \exp\left(\gamma\left(d_n^i + m\right) - \gamma \cdot d_p^j\right)} \tag{25}$$

where γ determines the largest scale of each similarity score. In the experiment, we set the γ as 10, because a large gamma might cause the result to become unstable. In addition, we set margin m as 0 by default. Instead of optimizing for a single pair, L_m assigns the different gradients to multiple pairs separately, which allows it to optimize multiple pairs simultaneously.

Algorithm 1 Training and Inference Process of the EMSCNet

Training:
Input: Train set $\boldsymbol{X}_{train}$; EMSCNet $G_b(.)$ with parameters θ_b^0; Dictionary encoder $G_k(.)$ with parameters θ_k^0; Dictionary $Dictionary(.)$; Multi-Head Contrastive Block MH(., .); Training epoch $epoch$;
Output: The optimized EMSCNet model $G_b(.)$ with parameters θ_b^{epoch};
for $i = 1$ to $epoch$ **do**
    $\boldsymbol{V} \leftarrow G_b(\boldsymbol{X}_{train})$;
    $\boldsymbol{V}_n, \boldsymbol{V}_p \leftarrow Dictionary(\boldsymbol{V})$;
    $\boldsymbol{V}_n^{\text{MH}} \leftarrow \text{MH}(\boldsymbol{V}, \boldsymbol{V}_n)$;
    $\boldsymbol{V}_p^{\text{MH}} \leftarrow \text{MH}(\boldsymbol{V}, \boldsymbol{V}_p)$;
    $\boldsymbol{V}_m \leftarrow \boldsymbol{V}_n^{\text{MH}} + \boldsymbol{V}_p^{\text{MH}} + \boldsymbol{V}$;
    $\text{predict}_1 \leftarrow softmax(\text{FC}_1(\boldsymbol{V}))$;
    $\text{predict}_2 \leftarrow softmax(\text{FC}_2(\boldsymbol{V}_m))$;
    Compute the L_{ce1} using Eq. (17);
    Compute the L_{ce2} using Eq. (18);
    Compute the L_m using Eq. (23);
    Compute the L_{kd} using Eq. (26)-(28);
    Compute the L_{total} using Eq. (29);
    Optimize $G_b(.)$ parameters θ_b^i with L_{total};
    Update $Dictionary(.)$ by $G_k(\boldsymbol{X}_{train})$;
    Optimize $G_k(.)$ parameters θ_k^i by Eq. (5);
end for
return θ_b^{epoch};
Inference:
Input: Test set $\boldsymbol{X}_{test}$; EMSCNet $G_b(.)$ without contrastive module; Parameters θ_b^{epoch};
Output: The predicted label predict_1;
$\boldsymbol{V} \leftarrow G_b(\boldsymbol{X}_{test})$;
$\text{predict}_1 \leftarrow softmax(\text{FC}_1(\boldsymbol{V}))$;
return predict_1;

3) Knowledge Distillation Loss: In the inference stage, the contrastive module needs to retain training samples for interaction with test samples. To save additional dictionary overhead and computation of the contrastive module, we remove the contrastive module. However, there is a large amount of available knowledge in the contrastive module. Instead of removing the module directly, we transfer the knowledge of the contrastive module to the backbone network classifier through knowledge distillation during training, and remove this module during inference. Specifically, the student and teacher probabilities are smoothed by a softmax function with temperature τ, and a higher value of τ produces a softer prediction distribution. The kth element of $\boldsymbol{p}_{\text{stu}}^{\tau}$ and $\boldsymbol{p}_{\text{tea}}^{\tau}$ are written as follows:

$$p_{\text{stu}}^{\tau}(k) = \frac{\exp\left(\text{FC}_1(\boldsymbol{v}_i)^k / \tau\right)}{\sum_{j=1}^{M} \exp\left(\text{FC}_1(\boldsymbol{v}_i)^k / \tau\right)} \tag{26}$$

$$p_{\text{tea}}^{\tau}(k) = \frac{\exp\left(\text{FC}_2\left(\boldsymbol{v}_{m,i}\right)^k / \tau\right)}{\sum_{j=1}^{M} \exp\left(\text{FC}_2\left(\boldsymbol{v}_{m,i}\right)^k / \tau\right)}. \tag{27}$$

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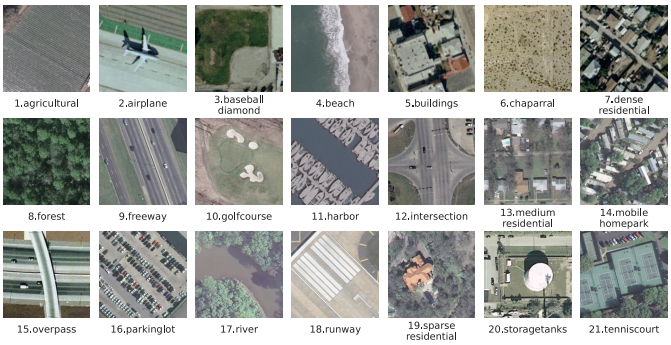


Fig. 4. Sample images for each category on the UCM dataset.

The knowledge distillation loss adopts the Kullback–Leibler (KL) divergence. It can be formulated as follows:

$$L_{\mathrm{kd}} = \tau^2 \cdot L_{\mathrm{KL}}\left(\boldsymbol{p}_{\mathrm{stu}}^{\tau} \parallel \boldsymbol{p}_{\mathrm{tea}}^{\tau}\right) = \tau^2 \cdot \sum_{i=1}^{M} p_{\mathrm{tea}}^{\tau}(i) \log \frac{p_{\mathrm{tea}}^{\tau}(i)}{p_{\mathrm{stu}}^{\tau}(i)}. \tag{28}$$

The total loss can be formulated as follows:

$$L_{\mathrm{total}} = L_{\mathrm{ce1}} + L_{\mathrm{ce2}} + \alpha L_m + \lambda L_{\mathrm{kd}} \tag{29}$$

where α and λ represent the loss coefficients that balance the importance of L_m and L_{kd}. α and λ are set as 1 by default.

Algorithm 1 provides the pseudocode for EMSCNet training and inference.

IV. Experiment and Analysis

To evaluate the effectiveness of the proposed EMSCNet, we perform extensive experiments on three public remote sensing datasets, including UC merced land-use (UCM) dataset [1], aerial image (AID) dataset [2], and NWPU-RESISC45 (NWPU) dataset [3].

A. Datasets

1) UCM Dataset: The UCM dataset is one of the most widely used datasets in the RSISC, where the images were collected from the United States Geological Survey National Map, Urban Area Imagery. It has 21 categories (see Fig. 4) and 1000 images in each category. Each image has three bands and a spatial size of 256 × 256.

2) AID Dataset: The AID dataset was collected by Wuhan University on Google Earth Image, and published in 2016. The AID is a high spatial resolution multisource dataset with a spatial size of 600 × 600. It contains 30 categories (see Fig. 5), and the number of images per category is not consistent. The dataset also contains three bands.

3) NWPU-RESISC45 Dataset: The NUPU dataset was collected by Northwestern Polytechnical University in 2016. It is a multisource dataset with 45 categories (see Fig. 6), and each category contains 700 images. Each image has a spatial size of 256 × 256.

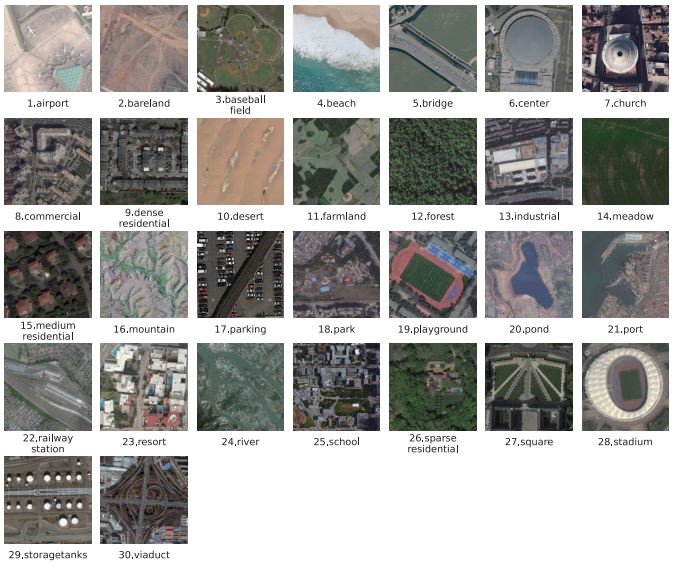


Fig. 5. Sample images for each category on the AID dataset.

B. Experimental Setup

We use Pytorch with a GeForce RTX 3090 for all experiments. The baseline ResNet and ViT are initialized with parameters pretrained on ImageNet [51], and other parts of EMSCNet are initialized randomly. The batch size is set as 64. We adopt different optimizer configurations for EMSCNet (ViT-B) and EMSCNet (ResNet-50) to achieve the best results. For EMSCNet (ResNet-50), the Adam [52] optimizer is used to train 80 epochs, the weight decay is 0.0005, and the learning rate is initialized to 0.0001, divided by 10 every 30 epochs. We use the SGD [53] optimizer to train EMSCNet (ViT-B) for 50 epochs. The corresponding momentum is set as 0.9, and the weight decay is 0.00005. The learning rate is initialized as 0.001, divided by 10 every 20 epochs.

We adopt the same data processing and evaluation metric as ET-GSNet [37]. The image augmentation methods include random flip and rotation. The input images are resized into 256 × 256. All images of the dataset are randomly divided into a training set and a testing set. The training ratios in UCM are set to 50% and 80%, respectively. The training ratios in AID are set to 20% and 50%, respectively. The training ratios in NWPU are set to 10% and 20%, respectively. All experiments are repeated five times, and we report the mean overall accuracy (OA) and standard deviation (STD), reflecting the overall performance of classification. In addition, we plot the confusion matrix, which can reflect the between-class classification errors and confusion degree.

C. Comparison With Some State-of-the-Art Methods

In Table I, we listed the SOTA method in recent years to compare with EMSCNet. ResNet-50 [9], ViT-B [33], GoogLeNet [2], and VGGNet-16 [2] are baseline methods. DCNN [10] and MSA-Network [54] are based on metric learning and attention mechanism, respectively. SCCov [55], CMFAM [56], MIDC-Net_CS [57], EMTCAL [35], and MG-CAP (Sqrt-E) [58] are multilayer and multigrained

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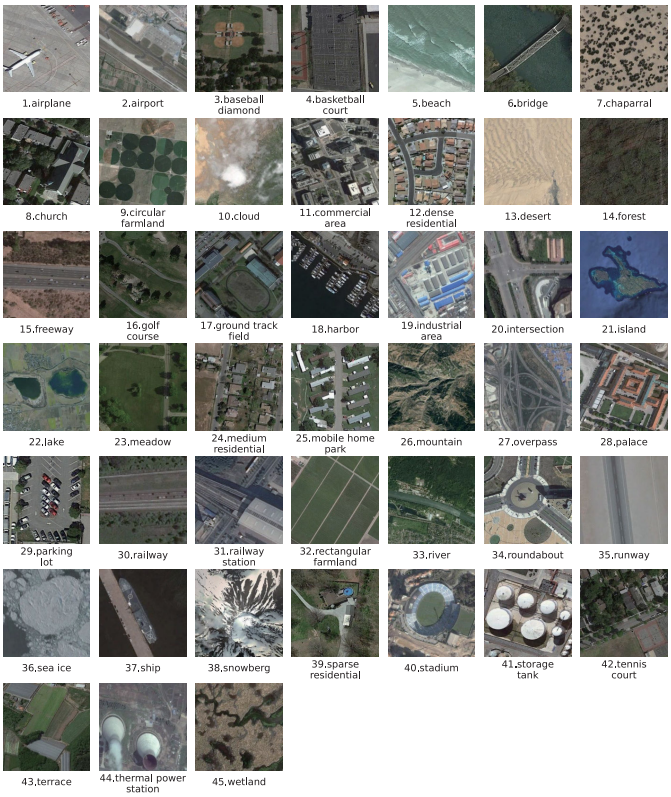


Fig. 6. Sample images for each category on the NWPU dataset.

feature methods. HHTL [59] and SCViT [36] are the improved methods for ViT. ET-GSNet [37] is a method based on knowledge distillation. In addition, these methods can be divided into CNN-based and ViT-based methods. We also implement the MLP-based method Cycle MLP [60].

1) Classification on the UCM Dataset: The results on the UCM dataset are shown in Table II. EMSCNet achieves the best accuracy on the UCM dataset. Specifically, with 50% training samples, EMSCNet can achieve 0.45% and 0.52% improvement over the baseline model when equipped with ResNet-50 and ViT-B. When 80% training samples is used, most methods can achieve 99% classification accuracy, and the classification accuracy tends to saturate. Our EMSCNet (ViT-B) can achieve a higher accuracy of 99.71% than the second-best method 99.57%. The results demonstrate the effectiveness of our proposed method on the small dataset.

2) Classification on the AID Dataset: Table III shows that our proposed EMSCNet achieves the best results in both CNN-based and ViT-based methods. When using 20% training samples, EMSCNet (ResNet-50) can achieve 95.13% accuracy with a 0.38% improvement over CMFAM [56], and EMSCNet (ViT-B) can achieve 96.02% accuracy with a 0.4% improvement over HHTL [59]. When using 50% training samples, EMSCNet (ResNet-50) can achieve 96.96% accuracy with a 1% improvement over the baseline, and EMSCNet (ViT-B) can achieve 97.35% accuracy with a 0.37% improvement over SCViT [36]. In most cases, EMSCNet exceeds the SOTA results. The result illustrates that our proposed contrastive module can achieve stable improvement on both CNNs and ViT backbones. With the increase in training data, the advantages of ViT-based methods are more prominent than CNN-based methods.

TABLE I

DETAILED TYPES AND PUBLICATION YEARS OF THE COMPARED METHODS. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, AND ∗: MLP-BASED METHODS)

Method	Year
GoogLeNet / VGGNet-16 [2] †	TGRS2017
ResNet-50 [9] †	CVPR2016
DCNN [10] †	TGRS2018
SCCov [55] †	TNNLS2019
CMFAM [56] †	GRSL2020
MIDC-Net_CS [57] †	TIP2020
MG-CAP (Sqrt-E) [58] †	TIP2020
MSA-Network [54] †	JSTARS2021
EMTCAL [35] †	TGRS2022
ViT-B [33] ‡	ICLR2021
ET-GSNet [37] ‡	TGRS2022
SCViT [36] ‡	TGRS2022
HHTL [59] ‡	JSTARS2022
Cycle MLP [60] ∗	ICLR2022
EMSCNet(ResNet-50) †	Ours
EMSCNet(ViT-B) ‡	Ours

TABLE II

COMPARISON OF OA AND STD ON THE UCM DATASET. THE BEST VALUES ARE MARKED IN BOLD. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, AND ∗: MLP-BASED METHODS)

Method	50% Training Ratio	80% Training Ratio
GoogLeNet [2] †	92.70 ± 0.60	94.31 ± 0.89
VGGNet-16 [2] †	94.14 ± 0.69	95.21 ± 1.20
ResNet-50 [9] †	98.25 ± 0.66	98.69 ± 0.49
DCNN [10] †	–	98.93 ± 0.10
SCCov [55] †	–	99.05 ± 0.25
CMFAM [56] †	98.34 ± 0.26	99.28 ± 0.21
MIDC-Net_CS [57]†	95.41 ± 0.40	97.40 ± 0.48
MG-CAP (Sqrt-E) [58]†	–	99.00 ± 0.10
MSA-Network [54] †	97.80 ± 0.33	98.96 ± 0.21
EMTCAL [35] †	98.67 ± 0.16	99.57 ± 0.28
ViT-B [33] ‡	98.58 ± 0.15	99.28 ± 0.21
ET-GSNet [37] ‡	–	99.29 ± 0.34
SCViT [36] ‡	98.90 ± 0.19	99.57 ± 0.31
HHTL [59] ‡	98.87 ± 0.28	99.48 ± 0.28
Cycle MLP [60] ∗	98.32 ± 0.28	99.05 ± 0.39
EMSCNet (ResNet-50) †	98.70 ± 0.46	99.44 ± 0.16
EMSCNet (ViT-B) ‡	$\mathbf{99.10 \pm 0.20}$	$\mathbf{99.71 \pm 0.17}$

3) Classification on the NWPU Dataset: The results on the NWPU dataset are shown in Table IV. We achieved the highest accuracy on the NWPU dataset. Specifically, when using 10% and 20% training samples, EMSCNet (ResNet-50) outperforms EMTCAL [35] method by 0.53% and 0.43%, respectively. Meanwhile, EMSCNet (ViT-B) is 0.86% and 0.63% higher than the second-best ViT-based method. The improvement of EMSCNet is obvious, because there are more serious intraclass diversity and interclass similarity in the large dataset, and our proposed method can effectively improve this problem.

D. Confusion Matrix Analysis

To intuitively evaluate the between-class errors, we provide confusion matrices for baseline ViT-B and EMSCNet (ViT-B) on the three datasets. As shown in Fig. 7(a) and (b), EMSCNet (ViT-B) achieves impressive performance on the UCM dataset with only one class misclassification. Compared

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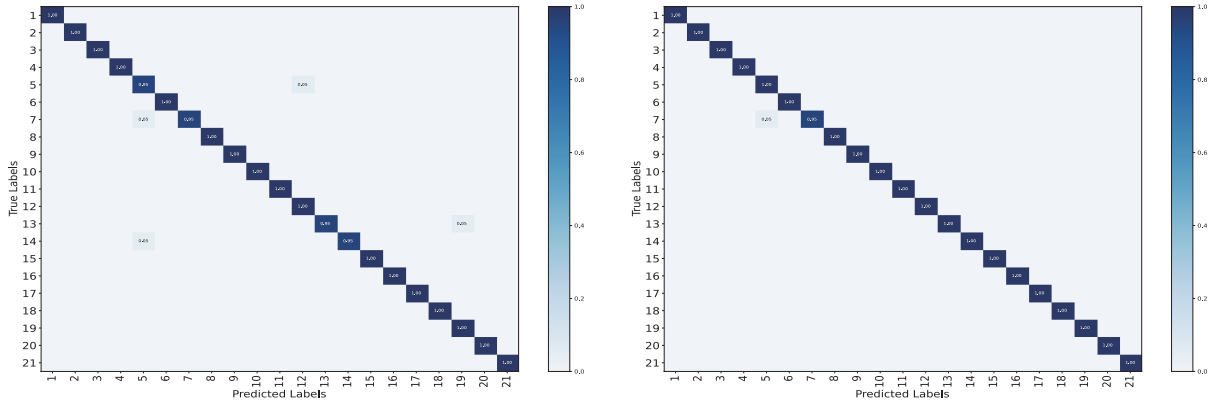


Fig. 7. Confusion matrices on the UCM dataset obtained by (a) ViT-B and (b) EMSCNet (ViT-B) models with a training ratio of 80%. The category numbers are consistent with Fig 4.

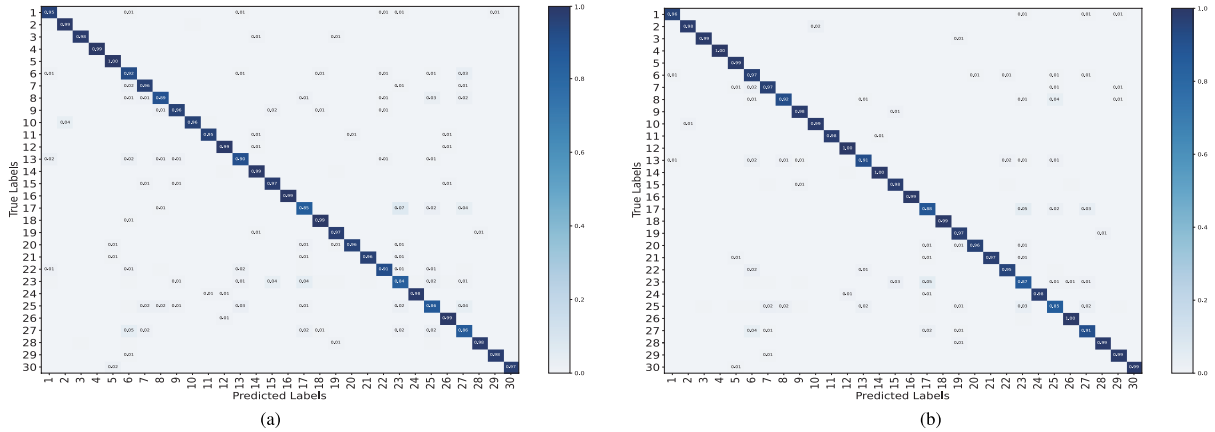


Fig. 8. Confusion matrices on the AID dataset obtained by (a) ViT-B and (b) EMSCNet (ViT-B) models with a training ratio of 20%. The category numbers are consistent with Fig 5.

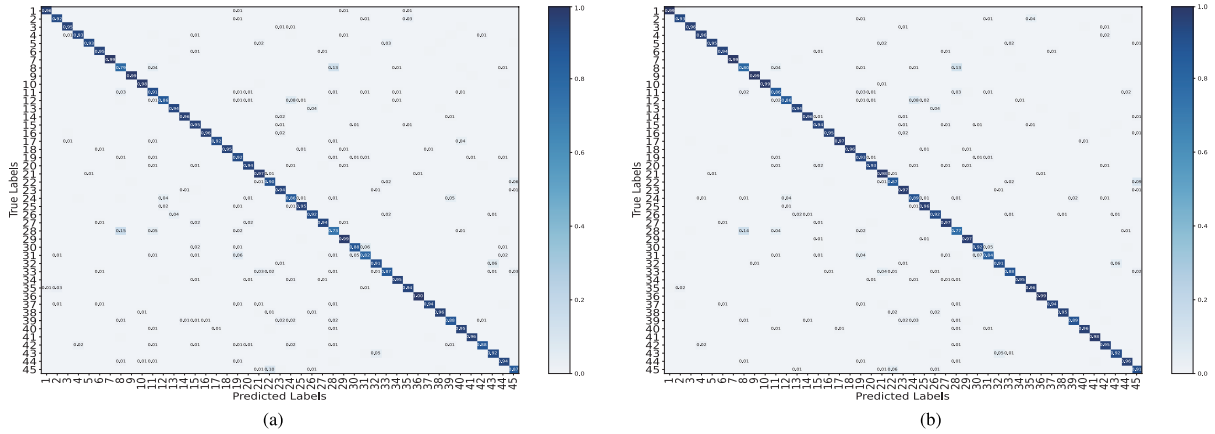


Fig. 9. Confusion matrices on the NWPU dataset obtained by (a) ViT-B and (b) EMSCNet (ViT-B) models with a training ratio of 10%. The category numbers are consistent with Fig 6.

to ViT-B, EMSCNet (ViT-B) has no confusion among the *medium residential* (13), and *mobile homepark* (14) categories, which illustrates that our method can improve the interclass similarity on the small dataset.

Fig. 8(a) and (b) show the confusion matrices obtained by ViT-B and EMSCNet (ViT-B) on the AID dataset, respectively. EMSCNet (ViT-B) have achieved steady improvements in the confusing categories of *center* (6), *commercial* (8), *desert* (10), *railway station* (22), and *resort* (23). The *desert* (10) misclassified as *bareland* (2) decreased from 0.04% to 0.01%, and the *center* (6) misclassified as *square* (27) decreased from 0.03% to 0.01%. This shows that our proposed method can improve the interclass similarity on the large dataset.

Fig. 9(a) and (b) show the confusion matrices obtained by ViT-B and EMSCNet (ViT-B) on the NWPU dataset,

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TABLE III

COMPARISON OF OA AND STD ON THE AID DATASET. THE BEST VALUES ARE MARKED IN BOLD. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, AND ∗: MLP-BASED METHODS)

Method	20% Training Ratio	50% Training Ratio
GoogLeNet [2] †	83.44 ± 0.40	86.39 ± 0.55
VGGNet-16 [2] †	86.59 ± 0.29	89.64 ± 0.36
ResNet-50 [9] †	92.57 ± 0.17	95.96 ± 0.17
DCNN [10] †	90.82 ± 0.16	96.89 ± 0.10
SCCov [55] †	93.12 ± 0.25	96.10 ± 0.16
CMFAM [56] †	94.75 ± 0.23	96.93 ± 0.16
MIDC-Net_CS [57] †	88.51 ± 0.41	92.95 ± 0.17
MG-CAP (Sqrt-E) [58] †	93.34 ± 0.18	96.12 ± 0.12
MSA-Network [54] †	93.53 ± 0.21	96.01 ± 0.43
EMTCAL [35] †	94.69 ± 0.14	96.41 ± 0.23
ViT-B [33] ‡	95.21 ± 0.16	96.96 ± 0.22
ET-GSNet [37] ‡	95.58 ± 0.18	96.88 ± 0.19
SCViT [36] ‡	95.56 ± 0.17	96.98 ± 0.16
HHTL [59] ‡	95.62 ± 0.13	96.88 ± 0.21
Cycle MLP [60] ∗	95.31 ± 0.15	96.87 ± 0.17
EMSCNet (ResNet-50) †	95.13 ± 0.10	96.96 ± 0.10
EMSCNet (ViT-B) ‡	$\mathbf{96.02 \pm 0.18}$	$\mathbf{97.35 \pm 0.17}$

TABLE IV

COMPARISON OF OA AND STD ON THE NWPU DATASET. THE BEST VALUES ARE MARKED IN BOLD. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, AND ∗: MLP-BASED METHODS)

Method	10% Training Ratio	20% Training Ratio
GoogLeNet [3] †	76.19 ± 0.38	78.48 ± 0.26
VGGNet-16 [3] †	76.47 ± 0.18	79.79 ± 0.15
ResNet-50 [9] †	88.48 ± 0.21	91.86 ± 0.19
DCNN [10] †	89.22 ± 0.50	91.89 ± 0.22
SCCov [55] †	89.30 ± 0.35	92.10 ± 0.25
CMFAM [56] †	91.08 ± 0.24	93.49 ± 0.17
MIDC-Net_CS [57] †	86.12 ± 0.29	87.99 ± 0.18
MG-CAP (Sqrt-E) [58] †	90.83 ± 0.12	92.95 ± 0.13
MSA-Network [54] †	90.38 ± 0.17	93.52 ± 0.21
EMTCAL [35] †	91.63 ± 0.19	93.65 ± 0.12
ViT-B [33] ‡	92.45 ± 0.12	94.74 ± 0.12
ET-GSNet [37] ‡	92.72 ± 0.28	94.50 ± 0.18
SCViT [36] ‡	92.72 ± 0.04	94.66 ± 0.10
HHTL [59] ‡	92.07 ± 0.44	94.21 ± 0.09
Cycle MLP [60] ∗	91.84 ± 0.17	94.27 ± 0.11
EMSCNet (ResNet-50) †	92.16 ± 0.07	94.08 ± 0.20
EMSCNet (ViT-B) ‡	$\mathbf{93.58 \pm 0.22}$	$\mathbf{95.37 \pm 0.07}$

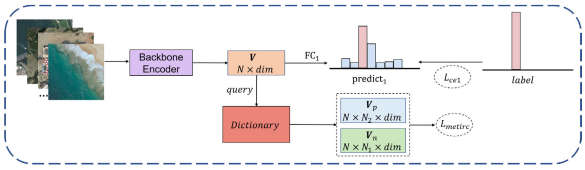


Fig. 10. EMSCNet with only metric loss retained. The L_{metric} can use multisample contrastive loss, contrastive loss, and triplet loss.

respectively. We achieved over 90% accuracy in 36 categories. The accuracy is increased from 88% to 95% on *tennis court* (42), from 96% to 99% on *airplane* (1), and from 88% to 92% on *railway* (30). This shows that our proposed method can improve the intraclass diversity.

E. Comparison of Different Metric Losses

To further verify the effectiveness of EMSCNet, as shown in Fig. 10, we only embed the multisample contrastive loss (23), contrastive loss (1), and triplet loss (3) before the last FC layer of the ViT-B, respectively. As shown in Fig. 11,

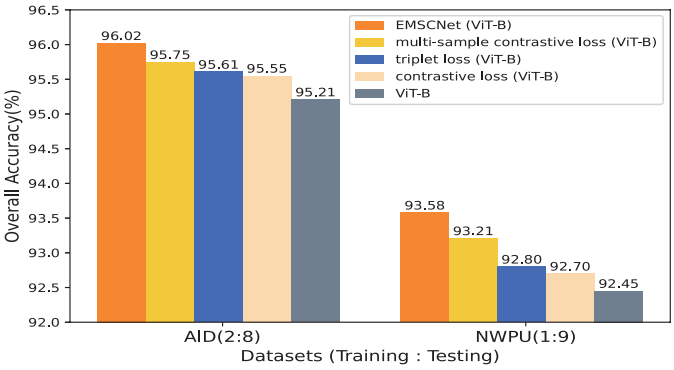
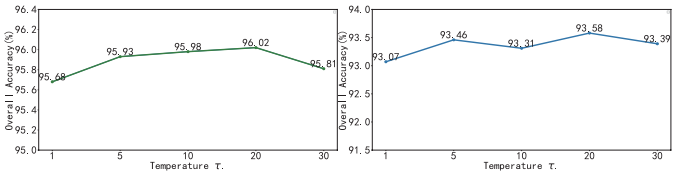


Fig. 11. Effect of different metric loss on the AID and NWPU datasets.



(a) (b)

Fig. 12. Effect of temperature τ for EMSCNet (ViT-B). τ on (a) AID dataset and (b) NWPU dataset.

multisample contrastive loss can achieve better results than contrastive loss and triplet loss, which illustrates the advantage of the multisample contrastive loss. In addition, EMSCNet (ViT-B) obtains obvious higher results than the others, which demonstrates the effectiveness of the multihead contrastive block.

F. Ablation Study

We carry out a detailed ablation study in this section. All experiments were performed with ViT-B on the AID and NWPU datasets under 20% and 10% training ratios, respectively.

1) Loss Function Parameter Sensitivity Analysis: We use KL divergence as the loss function of knowledge distillation, and the temperature τ controls the smoothness of the logits. As shown in Fig. 12, with the temperature τ increases, the accuracy increases first and then decreases. This is because a small τ causes the prediction distribution of the teacher tends to be hard label, while a large τ leads to small differences in the probability of different categories.

2) Ablation Study of Contrastive Module: The ablation results of the contrastive module are shown in Table V. When we remove the knowledge distillation loss L_{kd}, the OA drops by 0.24% and 0.48% on the AID and NWPU datasets, respectively. Moreover, when multisample contrastive loss is removed, the OA drops by 0.1% and 0.13% on the AID and NWPU datasets, respectively. Furthermore, when only keeping the multisample contrastive loss, the accuracy on the AID and NWPU datasets decreased by 0.27% and 0.37% compared to EMSCNet (ViT-B), which proves that the contrastive method through information interaction is superior to the metric learning method only using the distance function. In addition, during the inference stage, we retained a portion of the training samples to interact with the testing samples

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TABLE V

ABLATION RESULTS OF THE CONTRASTIVE MODULE ON THE AID AND NWPU DATASETS. (THE ✓ INDICATES THAT THE MODULE IS RETAINED)

L_m	L_{kd}	MH Contrastive Block	AID	NWPU
✓	✓	✓	96.02	93.58
✓			95.75	93.21
	✓	✓	95.92	93.45
✓		✓	95.78	93.10
			95.21	92.45

TABLE VI

ABLATION RESULTS OF THE CONTRASTIVE MODULE WITH VIT-B ON THE AID AND NWPU DATASETS UNDER DIFFERENT PROPORTIONS OF TRAINING SAMPLES. (THE † INDICATES THAT THE CONTRASTIVE MODULE IS RETAINED DURING THE INFERENCE STAGE)

Method	AID(20%)	AID(50%)	NWPU(10%)	NWPU(20%)
EMSCNet†	96.13	97.44	93.63	95.45
EMSCNet	96.02	97.35	93.58	95.37

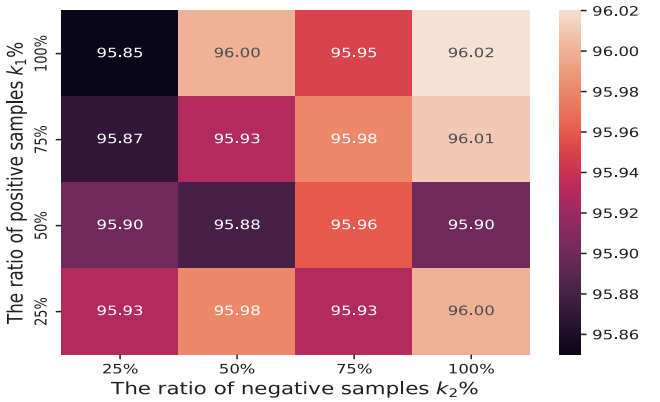


Fig. 13. Effect of the ratio of positive pairs k_1% and negative pairs k_1% for EMSCNet (ViT-B) on the AID dataset with a training ratio of 20%.

in the contrastive module. The final classification result was obtained by summing the outputs of the backbone and the contrastive module. The selection of positive samples during the inference stage may result in label leakage. To avoid this issue, we shared the parameters of the positive sample multihead contrastive block and the negative sample multihead contrastive block during the training stage. Subsequently, we randomly selected samples as contrastive samples during the inference stage. The experimental results are shown in Table VI, which demonstrates that retaining the contrastive module can achieve better results. However, considering the additional computational burden of the contrastive module, we remove the contrastive module by default in the inference stage.

3) Ablation Study of Dictionary: We verify the effect of the number of positive and negative pairs on the experimental results. As shown in Fig. 13, for each sample, the k_1% samples randomly selected in the same category are used as the queried positive samples, and the k_2% samples the randomly selected in other categories are used as the queried negative samples. When $k_1 = 100$ and $k_2 = 100$, we can achieve the best result on the AID dataset. In addition, when we only contrast one positive sample and one negative sample, the OA is only 95.73%. The results indicate that the similarity and discriminative knowledge can be obtained from multiple positive and negative pairs. As shown in Fig. 14, we also analyze the momentum update parameter m on the AID and NWPU datasets. As the momentum update parameter m decreases, the OA also decreases, indicating that feature inconsistency weakens the learning of the contrastive module. A low m might cause the model to collapse in unsupervised contrastive learning [15], but this does not happen in supervised learning. In addition, when the value of momentum coefficient m is 0, the parameters of encoders $G_k(\cdot)$ and $G_b(\cdot)$ are identical according to (5), so we can simplify the dictionary update process. The simplified EMSCNet replaces encoder $G_k(\cdot)$ with encoder $G_b(\cdot)$ to update the dynamic dictionary, which can greatly reduce the number of parameters and computational complexity.

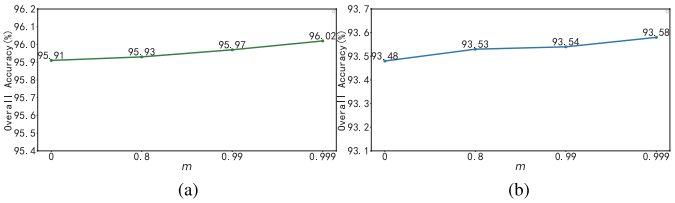


Fig. 14. Effect of momentum update parameter m for EMSCNet (ViT-B). m on (a) AID dataset and (b) NWPU dataset.

G. Number of Parameters and Computation Complexity

The computational complexity and parameters for one batch on the AID and NWPU datasets are shown in Table VII. Compared with the baseline model, EMSCNet (ResNet-50) and EMSCNet (ViT-B) need to use the momentum encoder $G_k(\cdot)$ to update the dictionary during training, and the $G_k(\cdot)$ will bring more parameters. The $G_k(\cdot)$ does not participate in backpropagation, so the amount of EMSCNet (ResNet-50) and EMSCNet (ViT-B) computational complexity is slightly increased compared to the baseline. The computational complexity and parameters of simplified EMSCNet (ResNet-50)† and EMSCNet (ViT-B)† are very close to the baseline model. At the same time, the OA is only about 0.1% lower than those of EMSCNet (ResNet-50) and EMSCNet (ViT-B). The result shows that our contrastive module does not significantly increase the complexity of the model.

H. Visualization Experiment

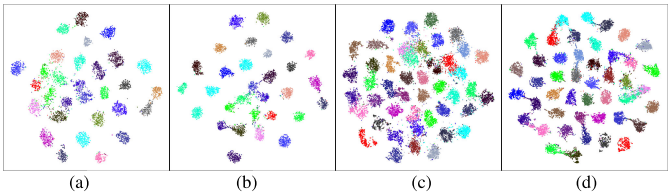
T-distributed random neighbor embedding [61] (t-SNE) is a nonlinear dimensionality reduction method that visualizes high-dimensional data by mapping each feature vector to a low-dimensional space. We visualize the classification results of ViT-B and EMSCNet (ViT-B) on the AID and NWPU datasets via t-SNE. As shown in Fig. 15, EMSCNet (ViT-B) has more concentrated cluster centers than ViT-B, and the distance between clusters is larger, which indicates that the feature vectors encoded by EMSCNet (ViT-B) have lower intraclass diversity and lower interclass similarity. Therefore, our

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TABLE VII

NUMBER OF PARAMETERS AND COMPUTATION COMPLEXITY FOR ONE BATCH ON THE AID AND NWPU DATASETS. (THE † INDICATES THE SIMPLIFIED EMSCNET MODEL WITH ENCODER $G_b(\cdot)$ INSTEAD OF $G_k(\cdot)$)

Method	Train Parameters	Train time	Test Parameters	Test time	AID	NWPU
VGGNet-16 [3]	15.75M	1.463s	15.75M	0.090s	86.59	76.47
ResNet-50 [9]	23.50M	0.885s	23.50M	0.250s	92.57	88.48
EMSCNet (ResNet-50)	50.69M	1.167s	23.50M	0.250s	95.15	92.24
EMSCNet (ResNet-50)†	27.49M	0.971s	23.50M	0.250s	95.10	92.16
Cycle MLP [60]	63.96M	1.452s	63.96M	0.287s	95.31	91.84
ViT-B [33]	85.63M	0.718s	85.63M	0.190s	95.21	92.45
EMSCNet (ViT-B)	173.64M	0.844s	85.63M	0.190s	96.02	93.58
EMSCNet (ViT-B)†	88.01M	0.750s	85.63M	0.190s	95.91	93.48



(a) (b) (c) (d)

Fig. 15. t-SNE results on the AID and NWPU datasets, where the training ratio is 20% and 10% of the dataset, respectively. (a) ViT-AID. (b) EMSCNet-AID. (c) ViT-NWPU. (d) EMSCNet-NWPU.

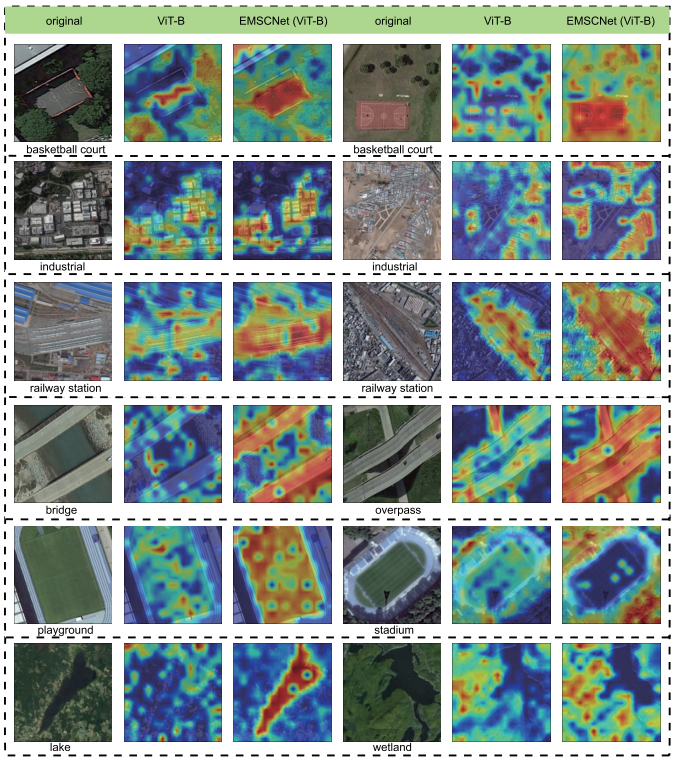


Fig. 16. Some samples of energy maps based on the fine-tuned ViT-B and the proposed EMSCNet (ViT-B).

proposed method can achieve better classification performance than the baseline.

Gard-CAM [62] is a popular visualization method which can intuitively reflect the location of interest in the network. Discriminative features are highly correlated with semantic labels. Therefore, locating the discriminative regions for RSISC is essential. We select some scene images to generate an energy map on the NWPU and AID datasets. As shown in Fig. 16, EMSCNet (ViT-B) classifies *basketball court*, *industrial*, and *railway station* by salient ground objects such as court, building, and railway, which shows that our proposed model can summarize the intraclass similarity knowledge. Both *bridge* and *overpass* categories contain bridges. EMSCNet (ViT-B) distinguishes them by ground objects under the overpass. The *playground* and *stadium* categories both contain a race track, and EMSCNet (ViT-B) distinguishes the two categories by the race track and surrounding context. *Lake* and *wetland* are easily confused. EMSCNet (ViT-B) distinguishes them by lake and land. The above example can prove that our proposed model can summarize the interclass discriminative knowledge. Furthermore, it can be observed from the energy map of the baseline model that ViT-B is deficient in both similarity and discriminative knowledge.

V. CONCLUSION

In this article, we design an EMSCNet model to address the problems of high intraclass diversity and interclass similarity in RSISC. Specifically, we propose a contrastive module consisting of a multihead contrastive block and a dynamic dictionary to summarize the discriminative and similarity knowledge in multiple sample pairs, and then transfer the dark knowledge of this module to the backbone with the knowledge distillation method. Furthermore, we verify that the impact of momentum m on supervised scene classification is much smaller than that of contrastive learning, which allows us further to reduce the computational complexity and parameters of EMSCNet. We conducted extensive experiments on three benchmark datasets. The results show that the discriminative and similarity knowledge of scene images can be summarized from the contrastive process of multiple positive and negative pairs. The contrastive module with information interaction can achieve better results than the metric learning only using distance loss. Knowledge distillation can effectively alleviate the negative impact of removing the contrastive module in the inference stage. Compared with previous SOTA methods, our proposed EMSCNet achieves more competitive results. In the future, we will further optimize the time consumption of EMSCNet and improve the performance on large-scale datasets.

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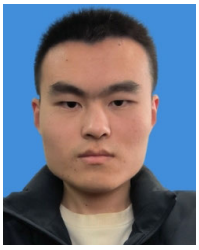
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